A Comprehensive Study of Bot Detection in Twitter: Evolution, Comparisons, and Challenges

Sheetal
Dept. Computer Science & Engg.
National Institute of Technology
Delhi, India
232211030@nitdelhi.ac.in

Amandeep Kaur
Dept. Computer Science & Engg.
National Institute of Technology
Delhi, India
amandeepk@nitdelhi.ac.in

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***Abstract*—The rise of automated social bots on Twitter threatens online discourse by spreading misinformation, amplifying propaganda, and manipulating public opinion. This review categorizes bot detection methods into Machine Learning (ML), Deep Learning (DL), Graph-based, Transformer-based, and Hybrid approaches. Traditional ML models, such as Decision Trees and SVMs, struggle with evolving bot behaviours, while DL techniques, including BiLSTMs, CNNs, and Transformers, capture complex patterns in text data. Pre-trained Language Models (PLMs) like BERT and RoBERTa further enhance detection through contextualized representations. Graph-based models analyse user interactions to identify bot clusters, and hybrid approaches improve robustness by integrating multiple methodologies. Moreover, This study presents benchmark dataset in this field along with the future directions. This study synthesizes research from 2019–2024, identifying challenges such as data imbalance and proposes future directions for advancing scalable, real-time bot detection frameworks.**

***Index Terms*—Social Media Bots, Bot Detection, Machine Learning, Deep Learning, Social Network Analysis.**

I. Introduction

Digital communication has changed due to the quick growth of online social networks, which allow for smooth interactions, quick information sharing, and worldwide connection. Twitter, Facebook, Instagram, and other social media platforms are important platforms for public engagement, political debate, marketing, and news sharing. But as our dependency on these networks has increased, so also have the cyber security threats, with social media bots emerging as a significant worry. With the use of scripts or Artificial Intelligence (AI) , these automated accounts may mimic human actions including sharing material, interacting with others, and starting conversations [1]. Some bots are used for detrimental objectives, such as disseminating false information, influencing public opinion, carrying out phishing attacks, and artificially magnifying particular narratives [2]. Some bots provide beneficial services like real-time alerts or customer support[1] and news aggregation, showing not all are malicious. A classification of malicious bots, highlighting their functionalities and risks, is illustrated in Fig. 1.

Researchers have investigated a number of detection methods, including Machine Learning (ML) [3], Deep Learning (DL), Graph Neural Network (GNN), Transformer-based models, and Hybrid approaches, in an effort to lessen the influence of Social bots. To differentiate Human users from bots, ML techniques employ classifiers such as decision trees, random forests [4], and support vector machines in conjunction with manually designed characteristics [5]. Ensemble techniques have been employed to further enhance bot detection performance [6]. Popular DL architectures [7] such as Convolutional Neural Networks (CNNs) [8] Recurrent Neural Network (RNN), [9] Long Short-Term Memory (LSTM) networks [10], Bidirectional LSTMs (BiLSTM) [11] [12], and Gated Recurrent Units (GRUs) have been widely applied in bot detection tasks [13]. DL models use neural networks to identify intricate patterns in behavioral and textual data [14]. Graph-based methods look for trends in bot behavior by examining user interaction networks [15] [16] [17] [18]. Transformer-based designs that capture temporal and contextual information, such as BERT and RoBERTa, have demonstrated excellent efficacy [19] [20]. Hybrid models combine many approaches to improve the resilience and accuracy of detection [21] [22]. Despite these developments, bot detection still faces difficul-

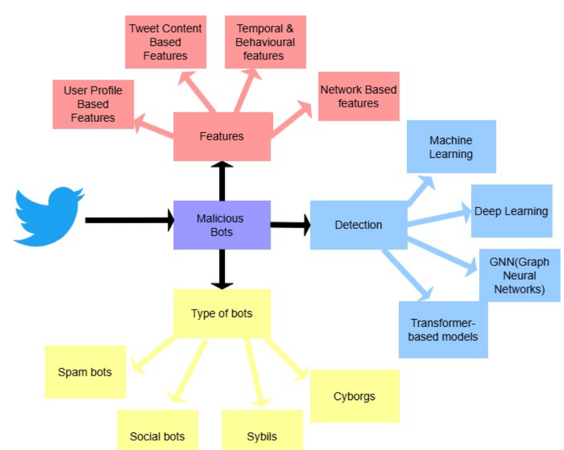

Fig. 1: Taxonomy of Malicious Bots

ties. Bots are always changing to avoid detection systems, and adversarial AI methods make things much more difficult. Furthermore, because many of the available datasets are out-of-

[1]https://twitter.com/earthquakesSF

TABLE I: Comparative Analysis of Bot Detection Approaches at various steps

Steps	Machine Learning (ML)	Deep Learning (DL)	Graph-based (GNNs)	Transformer-based & Hybrid Models
Data Collection	Twitter API , Web Scrapping, Tweepy [3]	Twitter API, Word Embeddings	Twitter API, Graph Networks	Pre-trained BERT, GPT, Text Embeddings [19]
Preprocessing	TF-IDF, Bag of Words (BoW), N-grams	Word2Vec, FastText, Glove, Embeddings	Node2Vec, Graph Sampling, MiniLM	WordPiece, Tokenization
Dim. Reduction	PCA, MinMax Scaling, LDA [19], Chi-Square	Autoencoders, T-SNE	Graph Attention Layers, Spectral Embeddings	Sentence Embeddings, Feature Pruning
Model Selection	RF, SVM, XGBoost [6], AdaBoost [2]	LSTM, CNN, BiLSTM, GRU, Self-Attention [8] [9]	GCN, GAT, GraphSAGE [16]	BERT, RoBERTa, DistilBERT, GPT [19], Bi-GRU+LSTM [22]
Classification	Decision Trees, SVM, KNN, Logistic Regression [1]	CNN-LSTM, RCNN, Attention-based Models [10]	Node Classification, Graph Clustering [17]	BERT-Classifier, RoBERTa-Classifier, GPT-Classifier [19] [20]
Evaluation	Accuracy, F1-score, Precision-Recall	ROC-AUC, F1-score, Precision-Recall	Graph Loss, Node-Level Accuracy, Graph Edit Distance	Perplexity, Cross-Entropy Loss [22]

date, domain-specific, or biased, access to high-quality labeled datasets is restricted. Class imbalances in datasets introduce an additional degree of complexity, even if feature selection and data pretreatment are essential for increasing detection accuracy. Important data for bot identification research is further limited by privacy issues, such as anonymized profiles and encrypted conversations.

This article analyzes bot detection at the account and tweet levels, highlighting research gaps, detection strategies, and their effectiveness. It explores advancements like multimodal fusion, adversarial training , and self-supervised learning to enhance social media security and trust.

II. Research Questions and Objectives

Bot detection has advanced from ML with handcrafted features to DL, GNNs, and transformers, enhancing accuracy and automation. However, real-time detection remains challenging due to computational limits and evolving bots. This study examines methodologies, effectiveness, and future improvements through the following questions:

- How have bot detection methodologies evolved from traditional ML models to advanced techniques like deep learning (DL), graph-based (GNNs), and transformer-based approaches?
- What are the key differences between ML, DL, GNN, and Transformer-based bot detection models in terms of performance and applicability?
- What challenges do bot detection models face in terms of accuracy, scalability, and robustness against adversarial bots?
- How do social network-based approaches (GNNs) differ from text-based approaches (BiLSTM, CNN) in bot detection?
- What are the major limitations of existing bot detection datasets, and how can future datasets be improved to enhance model generalization?

By addressing these questions, this study aims to provide insights into the strengths and weaknesses of various bot detection methodologies and suggest potential directions for future research.

III. Comparative Evolution of Bot Detection Techniques

Social bot detection has progressed through ML, DL, GNNs, and Transformer-based models. ML depends on feature engineering, DL captures complex patterns, GNNs exploit network structures, and transformers enhance text analysis. Hybrid models boost accuracy by combining these strengths, improving resilience against evolving bot tactics. Table II summarizes prior research, while Table I outlines detection methodologies and Figure 2 visualizes key techniques.

A. Analysis of Approaches

1) Traditional Machine Learning (ML) Techniques: Traditional ML methods detect bots by analyzing structured features from user activity, metadata, and linguistic patterns. [1]–[6].

Key Findings:

- RF [1], SVM [2], and XGBoost [6] are effective for bot detection.
- Ensemble learning boosts accuracy by integrating multiple classifiers [5].
- Feature selection techniques like PCA enhance classification [5].

Limitations:

- Relies on handcrafted features, limiting adaptability.
- Lacks continuous learning from new data.
- Struggles with deep contextual interpretation of bot-generated text.
- Word embeddings (Word2Vec, GloVe, FastText) enhance text representation.
- Attention mechanisms improve detection by handling obfuscated messages.

2) Deep Learning (DL) Techniques: DL improves bot detection by auto-extracting features from raw data, reducing manual effort. [7]–[14]. **Key Findings:**

  - RNNs [9], LSTMs [10], Bi-LSTM [11], and CNNs [12] effectively model sequential and spatial dependencies.

Limitations:

  - Requires significant GPU/TPU resources.

Paper ID	Models	Dataset Used	Analysis Domain	Input Features	Max Performance with Model	Future Direction
[1]	AdaBoost, Gradient Boosting, XGBoost	Kaggle dataset (2,797 Twitter accounts)	Account-Based Features	Verified status, friends count, followers count	89% (Gradient boosting with highest AUC score)	Tweet-based classification with enhanced features for improved performance.
[2]	One-class classifiers: Bagging-TPMiner, Bagging-RandomMiner, OC-KMeans, OC-SVM, Naïve Bayes	Cresci et al. dataset	Both Account-Based Features and Tweet based features	Account behavior features: retweet ratio, replies, favorites, hashtags, URLs, mentions, friend-follower ratio	0.89 AUC (One-Class SVM)	Integrate multi-class classifiers
[3]	Decision Tree, Random Forest, Naïve Bayes	Twitter API	Account-Based Features	Twitter account characteristics	93% (Decision Tree)	Explore more techniques and develop real-time detection.
[4]	Deep Learning, MLP, RF, Naïve Bayes, Rule-Based	11,000 tweets	Tweet-Based Features	special characters, sentiment features	77.15% (DL)	Improve interpretability
[5]	Random Forest, Adaboost, ANN, SVM, KNN	37,438 records	Both Account-Based Features and Tweet based features	Metadata: Followers count, account age, tweet frequency. Content: Text analysis, linguistic patterns	93% AUC	Improve accuracy, explore features, enhance detection models.
[6]	XGBoost, Graph Convolutional Networks (GCNs)	5.13M behavior records	Account-Based Detection	User features	Precision: 0.9217	Data augmentation, Model interpretability, Federated learning
[7]	JODIE, Random Forest	596,221 tweets	Tweet-Based Detection	User Interactions, Semantic Features	40.66% F1-Score improvement	Detect coordinated movements
[8]	Deep Boltzmann Machine	Botwiki-2019	Account-Based Features	10 key profile details	96.4% (Deep Boltzmann Machine)	CNN with attention for feature extraction
[9]	BiLSTM Network	Cresci dataset	Both Account-Based and tweet-based	Tweets + metadata (location, creation date)	97% (BiLSTM)	Improve accuracy, explore features, enhance detection models
[10]	ML Algorithms: SVM, DT, RF, AB, LR, k-NN DL Architecture: BiLSTM with attention	Cresci + Social Honeypot datasets	Both Account-Based and tweet-based	71 extracted features	Account: Accuracy: 0.9906, Tweet level: Accuracy: 0.8261	Explore transformers and GNNs for classification. Use Sentence-State LSTMs, explainability, GANs, and location-based features.
[11]	BiLSTM	Cresci-2017 dataset	Tweet-Based Features	Contextual content of tweets	Recall: 0.976 (Test Set 1) F-Measure: 0.926 (Test Set 2)	Extend to phishing detection
[12]	CNN, Text Classifier, Combined Classifier	Social Honeypot dataset and 1 KS to 10 KN dataset	Both Account-Based and tweet-based features	Text + metadata	F1-score: 99.68% (Social honeypot), F1-score: 93.12%(1 KS to 10 KN)	Classify spammers across social networks like Facebook and LinkedIn.
[13]	Bot-DenseNet, Multilingual Language Models	37,438 Twitter accounts	Account-Based Features	Metadata and text encoded via multilingual models	F1-Score:0.77 (RoBERTa)	Explores using GPT-3 and T5
[14]	RNN, CNN, PSO	8,574 Twitter accounts	Account-Based Features	Account Metadata	Accuracy: 99.2% (PSO)	Real-time adaptation, Advanced behavior analysis
[15]	Graph Convolutional Network (GCN), SVM, LSTM	PHEME dataset	Tweet + Metadata	User profile, Content, Interaction features	Accuracy: 76.3% (GCN)	Multi-class setup, Graph attention, OSINT tool
[16]	BotGCN, BotGAT, BotRGCN, SATAR, BotCL	Cresci-2015, TwiBot-20, TwiBot-22	Account-Based Features	Semantic, structural features	BotCL: 86.71%	Adaptive parameter search, efficient tuning, retweet-mention modeling, and heterogeneous graphs for bot detection.
[17]	BotRGCN	TwiBot-22	Both Account-Based and tweet-based features	User interactions (retweets, hashtags)	Accuracy: 79.07% (Follower relation), 86.28% (Co-Hashtag relation)	Explore new relations for bot detection
[18]	SStackGNN, GNNs + MLP	Cresci-15, TwiBot-20, MGTAB	Account-Based Features	User attributes, RoBERTa embeddings	Accuracy: 97.61%, F1: 97.42%	Enhance SStackGNN, Improve GNNs with new architectures
[19]	BERT, GPT-3, FNN	TweepFake, fox8-23	Tweet-Based Features	Contextual embeddings from PLMs	F1 Score: 93% (GPT-3)	Integrate datasets like TwiBot-20 and TwiBot-22
[20]	Bot-DenseNet (Deep Neural Network)	37,438 Twitter accounts	Account Based	Metadata (e.g., followers, activity) and text features (e.g., description, username)	Accuracy: 99.73% (BiGRU-LSTM)	Use Transformers to analyze bot-human embedding differences.
[21]	DeeProBot (Hybrid Deep Neural Network)	Botometer-labeled datasets	Both Account-Based and tweet-based features	AUC: 0.97	Real-time, adaptive bot detection.	
[22]	BiGRU, LSTM, GloVe embeddings	Cresci-2017, TwiBot-20	Tweet-Based Features	Semantic text embeddings (GloVe)	Accuracy: 99.73%, F1: 99.63%	Improve generalizability for diverse bots

TABLE II: Comparative analysis of different approaches

- Needs large-scale labeled datasets.
- Operates as a black-box, limiting explainability.

3) Graph-Based Neural Networks (GNNs): GNNs leverage social network structures to detect coordinated bot activities by capturing relationships between users rather than relying solely on individual features. Graph Convolutional Networks (GCN) propagate information across user connections, while Graph Attention Networks (GAT) enhance classification by assigning different importance scores to connections [15]–[18].

Key Findings:

- GCN and GAT improve bot detection accuracy [16].
- Node embeddings help identify bot clusters [17].
- Contrastive learning differentiates bot interactions from genuine users [18].

Limitations:

- High computational cost for graph construction.
- Cold start issue for new users with limited data.
- Vulnerable to adversarial manipulation.

4) Transformer-Based Approaches: Transformer models like BERT, RoBERTa, and GPT enhance bot detection using self-attention for contextual understanding. Fine-tuned BERT captures deep semantics, RoBERTa improves pre-training, and GPT analyzes tweet generation patterns. [19], [20]

Key Findings:

- Fine-tuned BERT models surpass traditional ML and DL models in accuracy [19].
- Context-aware classification improves bot differentiation [19].
- Pre-trained models accelerate training and reduce data dependency [20].

Limitations:

- High computational costs hinder scalability.
- Limited optimization for real-time detection.
- Potential biases affect multilingual performance.

5) Hybrid Approaches: Hybrid bot detection models integrate multiple techniques, such as ML, DL, GNNs, and transformers, to enhance detection accuracy and robustness. These models leverage the strengths of different architectures to address the limitations of individual approaches [21], [22].

Key Findings:

- deep neural networks with user profile features, achieving high accuracy in detecting social bots [21].
- Integrates BiGRU-LSTM and GloVe embeddings to effectively capture contextual patterns in bot behavior [22].

Limitations:

- Increased computational complexity.
- Higher resource requirements.
- Scalability challenges in real-time detection scenarios.

B. Trends, Challenges, and Research Gaps

The literature highlights several emerging trends:

- **Shift from ML to DL**: Traditional ML techniques are being replaced by deep learning models due to their ability to learn complex representations automatically.
- **Rise of Transformer Models**: Transformer-based architectures have gained traction due to their ability to capture contextual meaning in textual data [19].
- **GNN-based Approaches**: GNNs have gained traction for identifying bot networks by analyzing relationships between users [15] [16].
- **Hybrid Architectures**: Combining multiple methodologies, such as DL and GNN, has shown promising results in improving detection accuracy [20] [21].

Despite these advancements, several challenges remain:

- **Evolving Bot Strategies**: Bots continuously adapt their behavior, making it difficult to develop static detection models.
- **Data Limitations**: Many models rely on labeled datasets, which are often imbalanced or outdated.
- **Computational Constraints**: The high resource demand of DL and GNN-based techniques can hinder scalability.
- **Scalability Issues**: As social media data grows exponentially, scalable detection methods are needed.

IV. Datasets for Bot Detection Studies

The performance of bot detection models relies heavily on diverse, high-quality datasets. Researchers commonly use benchmark datasets containing real-world social media data, labeled bot accounts, and human profiles to evaluate ML, DL, graph-based, and hybrid approaches.[2]

A. Benchmark Datasets

A variety of datasets have been utilized for bot detection research, each offering unique characteristics for model evaluation:

- **Cresci-17** [2] [9] [10] [11] [13]: A widely used dataset containing labeled human and bot accounts from Twitter.
- **Varol-17** [20]: Includes a mix of human users, traditional bots, and social spambots.
- **Kaggle Twitter Bot Dataset** [5][3]: Features manually labeled bot and human accounts, frequently used for machine learning benchmarks.
- **Twibot-20** [7] [14] [15] [16] [18]: A large-scale dataset designed for deep learning-based bot detection, including profile, content, and network features.
- **Gilani-17** [21] : Collected using Twitter streaming data, with human-labeled bots.
- **Twibot-22** [16] [17]: An extension of Twibot-20 with richer graph-based features for analyzing social connections.
- **Cresci 2015** [16] [18]: CSV dataset with labeled real and fake Twitter accounts for bot detection.

[2]A comprehensive list of such datasets is available at:https://botometer.osome.iu.edu/bot-repository/datasets.html

[3]Dataset: https://www.kaggle.com/datasets/davidmartngutirrez/twitter-bots-accounts

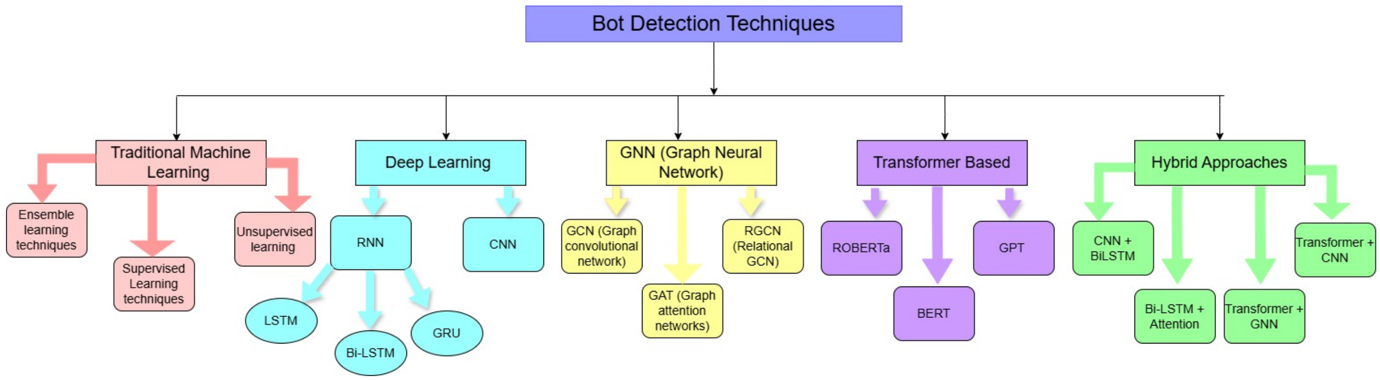

Fig. 2: Hierarchical Categorization of Bot Detection Methods

- **TweepFake** [19] : Contains fake tweets and bot-generated content for training transformer-based models.
- **BotWiki-2019** [8] : A dataset collected from the BotWiki platform, containing labeled bot accounts.

B. Challenges in Dataset Collection

Despite the availability of benchmark datasets, several challenges persist in bot detection research:

- **Data Imbalance:** Many datasets have a disproportionate number of human versus bot accounts, which impacts model generalization.
  To tackle class imbalance and improve model performance, researchers commonly use the following techniques:
  - **Over-sampling**: Increases minority class samples using duplication or synthetic methods like *SMOTE* (Synthetic Minority Over-sampling Technique) [10] [22].
  - **Under-sampling**: Reduces the majority class size to balance the dataset, although this may lead to loss of important information [10].
  - **Hybrid Sampling**: Combines over- and under-sampling to achieve better balance with minimal data loss [10].
- **Evolving Bot Behavior:** Bots continuously adapt, rendering older datasets less effective over time.
- **Privacy Constraints:** Social media data access is often restricted due to privacy regulations, limiting large-scale dataset collection.
- **Lack of Multilingual Data:** Most datasets are in English, restricting the application of models across different linguistic and regional contexts.

C. Future Directions in Dataset Development

Future research should focus on curating more comprehensive and up-to-date datasets with:

- **Real-time bot detection data streams**.
- **Cross-platform datasets** (e.g., Twitter, Facebook, TikTok, and LinkedIn).
- **Multimodal data** (text, images, and videos) for better bot detection.
- **Larger labeled datasets** incorporating adversarial bot behavior.

V. Discussion and Key Insights

Advancements in Twitter bot detection using ML, DL, GNNs, and hybrid models offer distinct strengths, yet real-time detection, transfer learning, and cross-platform analysis remain key underexplored areas for enhancing robustness.

A. Unresolved Challenges

- **Evolving Bot Strategies:** Many models fail to adapt to bots mimicking human behavior or using generative AI, making evolving adversarial bots a persistent challenge.
- **Generalizability:** Models trained on datasets like Cresci-2017 or Twibot-20 often underperform on unseen data, highlighting the need for domain-adaptive models for broader applicability.
- **Real-Time Detection:** Despite high accuracy, DL models are computationally intensive, limiting real-time use. Balancing efficiency and performance remains a key challenge.
- **Ethical and Privacy Concerns:** Automated bot detection risks false positives, making fairness, bias reduction, and user privacy critical concerns.

VI. Contributions and Future Directions

A. Key Insights

- **Model Trade-offs:** ML models are efficient but limited in detecting complex patterns. DL models, while effective in capturing richer features, require large datasets. GNNs excel at capturing network-level coordination, while Transformer-based models are strong in linguistic analysis. However, these models come at the cost of increased computational demands.
- **Fragmented Approaches:** Current methods often focus on isolated aspects—text, user networks, or behavior—lacking a unified approach for comprehensive bot detection.
- **Underexplored Challenges:** Issues such as data imbalance and multilingual adaptability remain inadequately addressed.

B. Future Research and Directions

- **Integration of Pretrained Language Models (PLMs)**: Leveraging transformers such as RoBERTa, T5, or GPT-based models could enhance contextual analysis and improve bot classification accuracy.
- **Graph Neural Networks (GNNs) and Heterogeneous Graph Models**: Advanced GNNs incorporating higher-order relationships and dynamic graph structures could enhance detection of coordinated bot activities.
- **Multimodal Bot Detection**: Combining text, images, metadata, and user interactions can improve detection capabilities by capturing multiple facets of bot behavior.
- **Self-Supervised and Few-Shot Learning**: Reducing dependence on labeled datasets by exploring self-supervised learning and few-shot learning approaches could enhance adaptability to new bot types.
- **Federated Learning for Privacy-Preserving Detection**: Developing decentralized bot detection models using federated learning could enable better privacy protection while maintaining model performance.
- **Explainable AI (XAI) for Transparency**: Ensuring interpretability in bot detection models can enhance trust and facilitate regulatory compliance, especially in sensitive applications like misinformation detection.

VII. Conclusion

This review examines the evolution of Twitter bot detection, from early rule-based and traditional ML models to advanced hybrid DL and transformer-based approaches. The integration of GNNs and PLMs like BERT and RoBERTa has improved detection accuracy by utilizing both contextual and network-based features. Despite these advancements, challenges remain in detecting sophisticated bots, real-time processing, and model generalization. Future research should focus on adaptive detection, multimodal analysis, and explainable AI to enhance trust and interpretability. As social media continues to shape information flow, effective bot detection is crucial for maintaining platform integrity and combating misinformation.

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